Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/08/2025
PI Name:	Delgado, Anca		
Project Title:	EFRI ELiS: Bioweathering Dynamics and Ecophysiology of Microbially Catalyzed Soil Genesis of Martian Regolith		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	National Science Foundation	**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	85287	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	09/01/2022	**End Date:**	08/31/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA HQ
Contact Monitor:	Koniges, Ursula	**Contact Phone:**	202-256-8786
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Flight Program:			
Flight Assignment:	This project is jointly sponsored by the National Science Foundation, Office of Emerging Frontiers and Multidisciplinary Activities (EFMA) and the National Aeronautics and Space Administration (NASA).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Aganaba, Timiebi (Arizona State University) Larson, Jean (Arizona State University) Tfaily, Malak (University of Arizona) Palmer, Andrew (Florida Institute of Technology)		
Grant/Contract No.:	IAA - National Science Foundation		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Emerging Frontiers in Research and Innovation (EFRI) Engineered Living Systems (ELiS) Abstract: A human mission to Mars is projected to take place by the late 2030s. Identifying viable sources of food will be critical to a successful mission given the significant distance between Earth and Mars. However, the Martian soil contains perchlorate salts at concentrations that are toxic to most forms of life including plants and crops. The overarching goal of this project is to explore the development and implementation of a microbially catalyzed soil genesis process that could remove perchlorate from Martian regolith while producing a soil containing the required organic carbon and inorganic nutrients to support plant growth and cultivation. To advance this goal, the project team proposes to investigate how to tune the bioweathering of Martian soil simulants by selected microbial consortia to generate viable soils that could support plant growth and crop cultivation on Mars. The successful completion of this project will benefit society through the generation of fundamental knowledge and the identification of microbial consortia that could convert sterile and toxic soils, such as the Martian regolith, to organic and nutrient rich soils that can sustain plant growth and crop cultivation. Additional benefits to society will be achieved through education and training including the mentoring of one postdoctoral researcher, one graduate student, and two undergraduate students at Arizona State University, one graduate student and one undergraduate student at the Florida Institute of Technology, and one graduate student at the University of Arizona. Mars regolith, or the surface material of Mars, is sterile with high concentrations of toxic perchlorate salts. In addition, the Mars regolith is devoid of soil organic matter (SOM) making plant cultivation for bio-regenerative life support and food production on Mars extremely challenging. The goal of this project is to design and develop scalable microbiological technologies and solutions to generate an organic and nutrient rich soil from Martian regolith. The guiding hypothesis of the proposed research is that the core metabolic processes of selected microbial consortia can be tuned to couple the production of a chemically diverse and stable SOM with the microbial reduction of perchlorate salts in Martian regolith. The specific objectives of the research include 1) experimental investigations of the extents and rates of perchlorate reduction by the selected microbial consortia under relevant Mars conditions using unsaturated and saturated regolith simulants, 2) characterization of the composition of the SOM and the bioavailable elements/nutrients that are generated from the microbial weathering of Mars regolith simulants using high resolution mass spectrometry coupled with liquid chromatography, and 3) demonstration of sustained seed germination and plant growth in the soil samples generated from the microbial weathering of Martian regolith simulants. The successful completion of this project has the potential for transformative impact through the generation of fundamental knowledge and the development of new methods to enable the conversion of sterile and toxic soils to organic and nutrient rich soils that could support plant growth and crop cultivation. To implement the education and outreach activities of the project, the Principal Investigators (PIs) propose to develop and implement a Research Experience for Teachers (RET) program to support the training of eight high school teachers at Arizona State University (ASU). In addition, the PIs plan to integrate the findings from this research into existing graduate/undergraduate courses and public outreach activities at their respective institutions including a yearly Open Door public outreach event at ASU and STEM podcasts of the Florida Institute of Technology NPR affiliate station with a focus on topics related to bioweathering, microorganisms for space exploration, and plant growth and crop cultivation in Martian regolith. This project is jointly sponsored by the National Science Foundation, Office of Emerging Frontiers and Multidisciplinary Activities (EFMA) and the National Aeronautics and Space Administration (NASA). This award reflects NSF's statutory mission and has been deemed worthy of support through evaluation using the Foundation's intellectual merit and broader impacts review criteria.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The successful completion of this project will benefit society through the generation of fundamental knowledge and the identification of microbial consortia that could convert sterile and toxic soils, such as the Martian regolith, to organic and nutrient rich soils that can sustain plant growth and crop cultivation. Additional benefits to society will be achieved through education and training including the mentoring of one postdoctoral researcher, one graduate student, and two undergraduate students at Arizona State University, one graduate student and one undergraduate student at the Florida Institute of Technology, and one graduate student at the University of Arizona.
Task Progress:	NOTE: Added to the NASA Task Book in September 2025. See Bibliography for publications to date (Ed., 9/8/25).
Bibliography Type:	Description: (Last Updated: 09/08/2025)
Abstracts for Journals and Proceedings	Ernsberger, K. "Designing an iterative research kit exchange program for remote high school science (evaluation)." 2024 ASEE Pacific Southwest (PSW) Section Conference, Las Vegas, Nevada, April 18-20, 2024. Abstracts. 2024 ASEE Pacific Southwest (PSW) Section Conference, Las Vegas, Nevada, April 18-20, 2024. Paper ID #44706. https://doi.org/10.18260/1-2--46029 , Apr-2024
Articles in Peer-reviewed Journals	Fackrell LE, Humphrey S, Loureiro R, Palmer AG, Long-Fox J. "Overview and recommendations for research on plants and microbes in regolith-based agriculture." npj Sustainable Agriculture. 2024 Sep 12;2(1):15. https://doi.org/10.1038/s44264-024-00013-5 , Sep-2024